Nasality and Voicing

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Abstract

Nasality and voicing have traditionally been treated as distinct phonological categories. The Unified Voicing and Nasality Hypothesis (UVNH) posits that both are phonetic manifestations of a single representational unit. Cross-linguistic phonetic and phonological research has shown that nasality and voicing frequently co-occur and interact in processes such as postnasal voicing assimilation, spontaneous prenasalisation, and intervocalic velar nasalization. Such phenomena are observed in diverse languages including Japanese, Greek, Asháninka, and Rotokas. Phonetically, a shared perceptual basis has been suggested in that voicing and nasality both enhance the low-frequency energy in the acoustic signal.

The chapter examines proposals regarding a unified prime across theoretical models, particularly Element Theory and Dependency Phonology, where competing proposals have been made regarding the roles of headedness and structure in encoding voicing and nasality. A variety phonological systems and typological data are analysed to assess the implications of the UVNH, including an apparent asymmetry in the distribution of true voicing and primary nasals. The aerodynamic and acoustic compatibility of voicing and nasality as well as potential counterevidence from nasal fricatives are also discussed. Finally, the discussion extends to the representation of nasality and voicing across different frameworks, highlighting the potential for cross-theoretical convergence, with the UVNH offering a parsimonious account of the phonological behaviour of nasality and voicing, with implications for mental representations, phonological typology, and the phonetics-phonology interface.

Keywords: voicing, nasality, representation, features, elements

1 Overview

In the early 1990s, a notable development in phonological studies aimed to constrain the generative potential of segmental expression by streamlining the representation of phonological features (e.g. Harris & Lindsey 1995, Nasukawa 1998, Ploch 1999). This involved a reduction in the number of features, for example through the amalgamation of two phonologically correlated features into a singular entity. This trend was particularly evident in a feature representation model known as Element Theory (ET).

Within the framework of Element Theory (Kaye et al. 1985, Scheer, 1999, Backley 2011; see also the chapter *Elements for Segmental Contrasts* and references therein), a specific endeavour was made to represent nasality (denoted by |N|) and voicing (denoted by |L|) through the utilization of a unified element. In other words, it was proposed to integrate these two phonological features, traditionally considered distinct, into a single representational unit within the Element Theory model. We refer to this proposal as the Unified Voicing and Nasality Hypothesis (UVNH).

Outside of ET, an approach to the conceptualization of grouping nasality and voicing under one category reminiscent in many ways of the UVNH is found in Kuroda (2002). In a distinct formulation of Feature Geometry, he contends that nasality should be contingent upon the node VCDVBR, which represents closed vocal folds.

The proposal that nasality and voicing are manifestations of a singular feature is ascribed to the recognition of phonological correlations between these properties. For instance, a phonological illustration of the correlation between the two properties is discernible in instances of *postnasal voicing assimilation*, which forces an obstruent following a nasal to share the voicing of the nasal, as amply illustrated in the literature (e.g. Itô & Mester 1986, Nasukawa 2000, Botma 2004). This assimilative process is observed across many phylogenetically and geographically diverse languages such as Asháninka, Yamato Japanese, Greek, and Zoque.

Further substantiating the nasal-voice correlation, additional instances manifest in processes such as *voiced obstruent prenasalisation*, which is observed in languages such as Northern Tohoku Japanese, various languages within the Reef Island–Santa Cruz family, a number of Pacific languages, and several Bantu languages. Another illustrative case is found in the occurrence of intervocalic *voiced-velar-obstruent nasalization* in conservative Tokyo Japanese. These cross-linguistic examples contribute to the comprehensive understanding of the interconnected nature of nasality and voicing.

In this chapter we focus mainly on the data supporting the UVNH, split across phonetic evidence (Section 2.1), evidence from phonological processes (Section 2.2) and evidence from phonological systems and their distributions (Section 2.3), before briefly illustrating and discussing a range of theoretical proposals that have been made with respect to the UVNH.

2 Data

2.1 Phonetics

2.1.1 Evidence in favour of the UVNH

The principal phonetic argument for the UVNH comes from the spectral acoustic effects of voicing and nasality. The vocal fold vibration of voicing introduces low-frequency energy into the signal in the form of a low-frequency buzz (often called the ‘voice bar’), typically somewhere in the 100-250 Hz region. Nasality introduces additional resonant frequencies and boosts especially those in the lower region of the spectrum, while also attenuating higher frequencies, thus functioning to some degree as a low-pass filter and amplifying the amount of low-frequency energy present in the signal, resulting in an acoustic signature commonly known as ‘nasal murmur’. The two signatures are illustrated in the spectral slices showing the mid-section of the hold phase of the pairs /b, m/, /d, n/ and /g, ŋ/ in Figure 1. Based on the element-theoretic assumption that the headedness of phonological elements is linked to the acoustic salience of an associated cue, it has been argued that nasality may thus represent a stronger, more salient exposition of a low-frequency cue associated with both voicing and nasality (Breit 2013, 2017).

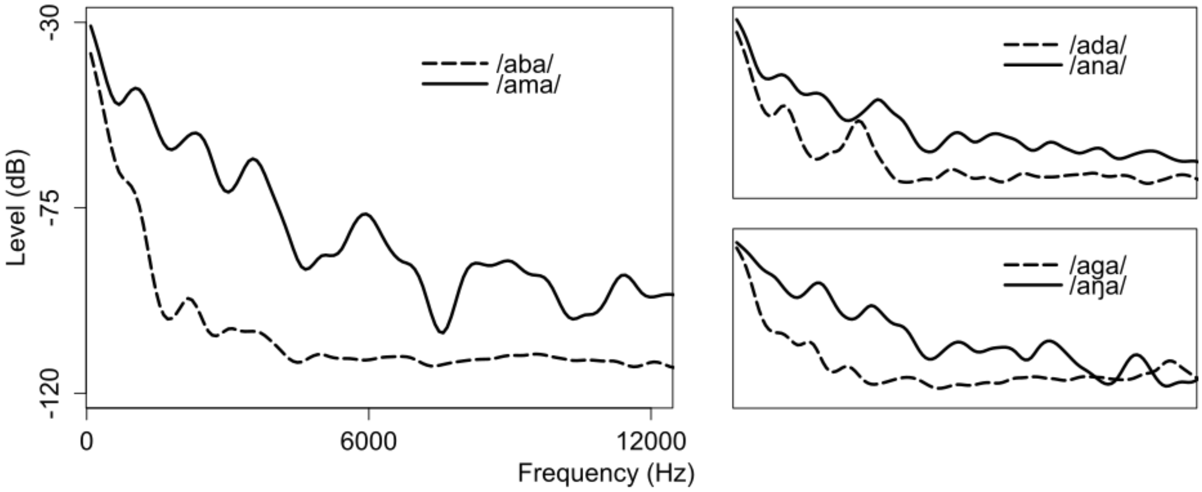


Figure 1: Spectral slices of the mid hold phase of voiced oral (dashed) and nasal (solid) stops at various places of articulation (from Breit 2017:17)

The phonetic realisation of voiceless (or fortis) nasals has been cited as another piece of phonetic evidence for the UVNH. It is widely agreed that these types of nasals are realised in at least two distinct ways: in the first type of languages, such as Burmese, Mizo and Amuzgo[1], the initial portion of a voiceless nasal is voiceless aspirated, followed by a modal voiced nasal period (viz. [m̥ʰm]), while in the second type of languages, such as Angami, Tibetan and Xumi, an initial voiceless nasal portion is followed by a period of nasal aspiration, which is at least partially voiced (e.g. Dantsuji 1984, Bhaskararao & Ladefoged 1991, Chirkova, Basset & Amelot 2019). Note however that the presence of voicing in the second type is highly variable, being quite prominent in Tibetan but minimal in Xumi, as shown by Chirkova, Basset & Amelot (2019). A third type, where an initial modal voiced nasal period is followed by a period of voiceless nasalised aspiration (viz. [mm̥h̃]) may be exemplified in Sumi (Harris 2010), Sukuma (Maddieson 1991) and Welsh (Scully 1973, Ball and Williams 2000). There again appears to be a degree of variation in terms of devoicing, to the extent that e.g. in Welsh it appears to frequently be the case that the ‘voiceless’ nasal is actually phonetically voiced for its entire duration (viz. [mɦ̃]), see e.g. Breit (2017:35). What is common to all these various realisations of voiceless/fortis nasals is that they retain some phonetic voicing, despite frequently contrasting with a voiced/lenis series and/or patterning phonologically with other voiceless/fortis segments.

It has been argued that the retention of partial voicing in such nasals principally serves to enhance the perceptibility of the nasal’s place cues, which are extremely weak in purely voiceless nasal airflow (Ohala 1975, Dantsuji 1984). However, this explanation may appear somewhat wanting at least in the case of voiceless nasals of the latter two types, where the offset of the nasal is characterised by significant oral airflow.

Proponents of the UVNH have argued that the presence of voicing in such fortis nasals is consistent with the assumption that voicing is an inherent property of nasality. If this is the case, fortis nasals must underlyingly involve both a primitive encoding voicing/nasality (viz. an amalgamated [+voice, +nasal]-feature, or |L|) and a primitive encoding fortisity/aspiration (viz. [+spread glottis], or |H|), since the voicing feature cannot be decoupled from nasality phonologically if these are encoded by the same feature. This predicts that the phonetic component is left with the task of realising conflicting cues (voicing and

[1] In Amuzgo, these voiceless nasals might be derived from underlying /hN/ sequences, though they appear to be contrastive on the surface, see e.g. Herrera 2000.

voicelessness/aspiration) for such segments, which is resolved by a language-specific sequential encoding, giving rise to the different types of 'voiceless' nasals discussed above (Botma 2004; Breit 2013, 2017).[2]

2.1.2 Potential counterevidence to the UVNH

It has been argued widely that nasalization is, to at least some degree, aerodynamically incompatible with (buccal) frication (e.g. Ohala 1975, Ohala and Ohala 1993), principally because the leakage of air through an open velopharyngeal port in nasalized segments prevents the build-up of sufficient oral air pressure to produce frication in the oral cavity. Ohala (1975) argues that, if this argument is correct, nasalization would be expected to have a particularly detrimental effect on voiced fricatives, as the additional maintenance of a sufficient pressure differential across the glottis together with leakage through the velopharyngeal port and the build-up of oral pressure to produce frication would overall likely be too demanding. This incompatibility of voicing and nasality seems to be corroborated by later experimental evidence from Ohala, Solé & Ying (1998), who showed that the effect of velopharyngeal leakage was more detrimental to the production of voiced than voiceless oral fricatives. However, the evidence appears to be much less clear cut than suggested by Ohala and colleagues, both regarding the status of nasalized oral fricatives more widely (see Shosted 2006 for a comprehensive overview), and with respect to voiced oral fricatives more specifically. For instance, investigating nasal harmony systems, Walker (2000) found that voiced fricatives are more compatible with nasality than voiceless obstruents. In any case, this potential aerodynamic antagonism is limited specifically to nasalized oral/buccal fricatives, which are cross-linguistically rare and whose status and realization are frequently surrounded by debate. There are no known physiological or articulatory factors linking the production of voiced and nasal segments more generally, and a physiological connection seems unlikely given that the (non-linguistic) default states of both the velopharyngeal port and the glottis are open. Instead, the principal phonetic connection between voicing and nasality appears to be acoustic in nature.

2.2 Processes

2.2.1 Postnasal voicing

Phonological studies reveal significant correlations between nasals and voiced obstruents. A paradigmatic example of this relationship is observed in postnasal voicing assimilation, which is evident across a diverse set of languages, including Gaelic (Celtic, Scotland; Oftedal 1956), Greek (Hellenic, Greece; Newton 1972), Asháninka (Arawak, Peru; Dirks 1953), Jicaque (Hokan, Honduras; Campbell & Oltrogge 1980), Puyo-Pongo Kichwa (Quechuan, Ecuador; Rice 1993), Zoque (Mixe-Zoquean, Mexico; Herrera 1995), Kikuyu and Ndebele (Bantu, Kenya and South Africa, respectively; Herbert 1986), Kpelle (Mande, Liberia; Welmers 1962), Mbiywom and Oykangand (Paman, Australia; Hale 1976, Sommer 1969), Wembawemba (Kulinic, Australia; Hercus 1986), and Yamato Japanese (Japonic, Japan; Itô & Mester 1986). As an illustration, Asháninka features a distributional constraint where an initial obstruent following a nasal must share the voicing of the nasal, as exemplified in (1).

[2] This likewise makes the relatively strong prediction that only aspiration languages, where an aspiration feature such as [+sg] or |H| is phonologically active, can encode a fortis/lenis contrast in nasals. This observation has also been made outside of element-theoretic work by Lombardi (1994).

(1) Asháninka (Dirks 1953)

kombiróši	‘palm leaf’	*kompiróši
nihánda	‘far away’	*nihánta
nišíndʲo	‘my daughter’	*nišíntʲo
kiríŋga	‘downstream’	*kiríŋka

Another instance of this phenomenon is observed in Greek. When a nasal is followed by a stop in Greek (Newton 1972: 93–99), the latter undergoes postnasal voicing.

(2) Greek (Newton 1972: 93–99)

/áNtras/	‘man’	[ándras]	(or [ádras])
/éNporos/	‘merchant’	[émboros]	(or[éboros])
/sinkenís/	‘relative’	[siɲɟenís]	(or [siɟenís])

As shown in (2), except in loanwords, stops preceded by a nasal are considered to be lexically voiceless but undergo phonetic voicing. Postnasal voicing in Greek can also be observed to occur across boundaries, e.g. /ton patéra/ ‘the.M.SG.ACC father’ → [tombatéra], /tin kiría/ ‘the.F.SG.ACC lady’ → [tiŋgiría] (see also Kainada 2009).[3]

Postnasal voicing is also observed in Puyo-Pongo Kichwa. As discussed by Rice (1993: 315) and Botma (2004: 32), this language exhibits phonologically active nasals and displays a process that operates exclusively in derived environments. The examples in (3) indicate that the lexical forms of the objective (OBJ), genitive (GEN) and locative (LOC) suffixes are /-ta/, /-pa/, and /-pi/. However, when these suffixes are attached to a stem ending with a nasal consonant, they manifest themselves as [-da], [-ba], and [-bi], respectively.

(3) Puyo-Pongo Kichwa (Rice 1993: 315, Botma 2004: 32)

/wasi-ta/	‘the others-OBJ’	[wakin-da]	‘the house- OBJ’
/sinik-pa/	‘porcupine-GEN’	[kam-ba]	‘you- GEN’
/saʧa-pi/	‘jungle-LOC’	[hatum-bi]	‘big one- LOC’

Postnasal voicing is additionally discernible in Yamato Japanese, a native class within the four lexical strata (Itô and Mester 1986, *et passim*)[4]. Similar to the previously mentioned examples, a nasal-obstruent cluster must be voiced in Yamato Japanese.

(4) Yamato Japanese

to**mb**o	‘dragon fly’	*tompo
ši**nd**oi	‘tired’	*šintoi
ka**ŋ**gae	‘thought’	*kaŋkae
ko**ŋ**gari	‘done to a golden brown’	*koŋkari

This phenomenon is found not only within lexical items, but also across a morpheme boundary. For example, verbal suffixes such as *-te*, *-ta*, *-tari*, and *-tara*, when attached to a stem ending with a nasal, show voicing assimilation.

[3] For further discussion of the rationale for analysing this as a process rather than a mere distributional fact, see Newton (1972) and Pagoni (1993).

[4] This postnasal voicing is not observed in the other three classes－Sino-Japanese, onomatopoeic, and loanwords within the four lexical strata.

For example, when the gerundive suffix /-te/ is attached to a stem ending with a vowel, there are no morphophonemic alternations; it is always realized as [te], as given in (5).

(5) Yamato Japanese: Stems ending with a vowel (Nasukawa 2005: 3)

/mi/	+	/-te/	→	[mi**t**e]	'see' (gerundive)
/ne/	+	/-te/	→	[ne**t**e]	'sleep' (gerundive)

On the other hand, when the suffix is attached to a stem ending with a consonant, there are morphophonemic alternations such as velar vocalization, gemination, *b*-nasalization, and *t*-voicing, as shown in (6).

(6) Stems ending with a consonant

a.	/tok/	+	/-te/	→	[toite]	'solve' (gerundive)
	/tat/	+	/-te/	→	[tatte]	'stand up' (gerundive)
	/tas/	+	/-te/	→	[taɕite]	'stand up' (gerundive)
b.	/to**b**/	+	/-te/	→	[ton**d**e]	'sharpen' (gerundive)
	/to**g**/	+	/-te/	→	[toi**d**e]	'sharpen' (gerundive)

Our current focus lies on the *t*-voicing phenomenon exhibited by the gerundive suffix /-te/. Specifically, while /-te/ remains intact when affixed to stems ending with a voiceless obstruent, the /t/ of the suffix undergoes a change, being realized as [d], when attached to stems ending with a voiced obstruent.

When it comes to sonorant consonants /r/ and /w/, which appear as stem-final segments, they are treated as vowel-like; no alternation is involved.

(7) Stems ending with a sonorant consonant

/tor/	+	/-te/	→	[totte]	'take' (gerundive)
/kaw/	+	/-te/	→	[katte]	'buy' (gerundive)

However, when the gerundive suffix /-te/ is preceded by nasal-final stems, it is realized as [de], as given in (8).

(8) Stems ending with a nasal

/si**n**/	+	/-te/	→	[ɕi**nd**e]	'die' (gerundive)
/ka**m**/	+	/-te/	→	[ka**nd**e]	'chew' (gerundive)

Given this, though typically classified as sonorants, nasals in Japanese are treated as voiced obstruents. Their voicing is considered to trigger post-nasal obstruent voicing[5].

An alternative analysis posits that the lexical form of the gerundive should be represented as /-de/ rather than /-te/. However, this perspective encounters challenges in elucidating the source of devoicing observed when the gerundive suffix is preceded by stems ending with a vowel, /r/, or /w/.

[5] However, note that this observation contradicts the patterns observed in Lyman's Law, which prohibits the occurrence of more than one voiced obstruent within a given word, including compounds. Lyman's Law categorizes nasals as sonorants rather than voiced obstruents, an inconsistency that would certainly warrant further discussion.

2.2.2 Spontaneous prenasalisation

A further correlation between nasal and voice is observed in *spontaneous prenasalisation*—the phenomenon that demonstrates alternations between voiced obstruents and prenasalized voiced plosives, with no discernible lexical indication of nasality in either the target position or its environment. This is found in some dialects of Japanese, some Western Indonesian languages and several Bantu languages. Examples from Northern Tohoku Japanese are given below (Nasukawa 2005: 6–7).

(9) Northern Tohoku Japanese

çi	+	darɯma	→	çi$^{\mathbf{n}}$**d**arɯma
‘fire’		‘Dharma’		‘be covered with flames’
tsɯɯ	+	gakkoo	→	tsɯ$^{\mathbf{ŋ}}$**g**akko
‘middle’		‘school’		‘junior high school’
o	+	baasaɴ	→	o$^{\mathbf{m}}$**b**asaɴ
(politeness)		‘grandmother’		‘grandmother (polite form)’
niɣɯ	+	ǰa^{ŋ}ga	→	niɣɯ$^{\mathbf{n}}$**ǰ**a^{ŋ}ga
‘meat’		‘potato’		(name of dish)

The instances in (9) serve as illustrations of the compounding process in Northern Tohoku Japanese. Under this process, when the initial segment of the second member of a compound is a voiced obstruent, it is realized as its prenasalised counterpart. In this case, nasality is not inherent lexically in either the target of the process or its surroundings. Importantly, the target must be in an intervocalic environment.

Indeed, this phenomenon is not limited to word formation, but it is also observed in the lexical distribution of prenasalised plosives. To illustrate, compare the forms in the Tokyo dialect in the left column of (10) with the counterparts in the northern Tohoku dialect on the right.

(10) Intervocalic voiced stop prenasalization (2005: 7)

	Tokyo Japanese	Northern Tohoku	
a.	ka**g**i	ka$^{\mathbf{ŋ}}$**g**i	‘key’
	sa**g**a	sa$^{\mathbf{ŋ}}$**g**a	‘destiny’
	i**g**ai**g**a	i$^{\mathbf{ŋ}}$**g**ai$^{\mathbf{ŋ}}$**g**a	‘thorny’
b.	ha**d**a	ha$^{\mathbf{n}}$**d**a	‘skin’
	ku**d**a	kɯ$^{\mathbf{n}}$**d**a	‘pipe, tube’
	i**d**o	i$^{\mathbf{n}}$**d**o	‘well’
c.	ka**b**u	ka$^{\mathbf{m}}$**b**ɯ	‘turnip’
	ki**b**a	ki$^{\mathbf{m}}$**b**a	‘tusks’
	sa**b**i	sa$^{\mathbf{m}}$**b**i	‘rust’

As indicated in (10), prenasalisation can be observed in all plosives occurring between vowels.

In contrast, the process does not affect voiced oral plosives found in alternative positions, and voiceless counterparts of plosives occurring between vowels remain unaffected. The illustration for this distinction is provided below.

(11) No voiceless stop prenasalization (Nasukawa 2005: 7)

	Tokyo Japanese	Northern Tohoku Japanese		
a.	**g**akkoo	gakkoo	*$^{\text{ŋ}}$**g**akkoo	‘school’
	darɯma	darɯma	*$^{\text{n}}$**d**arɯma	‘Dharma’
	baku	baɣɯ	*$^{\text{m}}$**b**akɯ	‘tapir’
b.	ka**k**i	ka**ɣ**i	*ka$^{\text{ŋ}}$**g**i, *ka$^{\text{ŋ}}$**k**i	‘persimmon’
	sa**k**a	sa**ɣ**a	*sa$^{\text{ŋ}}$**g**a, *sa$^{\text{ŋ}}$**k**a	‘slope’
	ha**t**a	ha**ɾ**a	*ha$^{\text{n}}$**d**a, *ha$^{\text{n}}$**t**a	‘flag’
	ka**t**a	ka**ɾ**a	*ka$^{\text{n}}$**d**a, *ka$^{\text{n}}$**t**a	‘shoulder’

As demonstrated in (11a), the initial position within a word does not permit the prenasalisation of voiced oral plosives. Furthermore, the category of segments susceptible to prenasalisation is confined specifically to voiced plosives. The contrast outlined in (11b) delineates that voiceless plosives situated between vowels do not undergo prenasalisation; rather, they are potential targets for subjects for vocalization.

2.2.3 Intervocalic velar nasalization

A comparable phenomenon has been documented in standard Japanese. Analogous to the occurrence of spontaneous prenasalisation, certain conservative Tokyo and northern Kanto dialects of Japanese manifest intervocalic velar nasalization, wherein a voiced velar obstruent undergoes complete nasalization. This is exemplified in (12).

(12) Intervocalic velar nasalization (Nasukawa 2005: 8)

šoo	+	gakkoo	→	šoo**ŋ**akkoo
‘small’		‘school’		‘primary school’
čuu	+	gakkoo	→	čuu**ŋ**akkoo
‘middle’		‘school’		‘junior high school’
tei	+	gi	→	tei**ŋ**i
‘to fix’		‘faithfulness’		‘definition’
kai	+	guɴ	→	kai**ŋ**uɴ
‘sea’		‘army’		‘navy’
šu	+	gei	→	šu**ŋ**ei
‘hand’		‘art’		‘handicraft’
kai	+	goo	→	kai**ŋ**oo
‘to meet’		‘to assemble’		‘meeting’

Within the morphosyntactic concatenation illustrated in (12), when the initial segment of the second element in a compound is a voiced velar obstruent, it is construed as its nasalized counterpart. Likewise, this occurrence is not contingent upon inherent nasality within the lexical entry; the critical criterion for the process to take effect is the placement of the target within an intervocalic context.

This intervocalic nasalization is evident in both under morphosyntactic concatenation and in the lexical distribution of velar nasals, as depicted in (13).

(13) The lexical distribution of velar nasals (Nasukawa 2005: 9)

Non-conservative Tokyo Japanese		Conservative Tokyo Japanese
sa**g**a	‘destiny’	sa**ŋ**a
ka**g**i	‘key’	ka**ŋ**i
i**g**ai**g**a	‘thorny’	i**ŋ**ai**ŋ**a
ki**g**u	‘tool’	ki**ŋ**u
ka**g**e	‘shadow’	ka**ŋ**e
ka**g**o	‘basket’	ka**ŋ**o

In contrast to the non-conservative Tokyo Japanese forms on the left, the conservative Tokyo Japanese forms on the right consistently lack voiced velar plosives between vowels. Instead, the corresponding sounds in this context are uniformly nasal.

2.3 Systems and Distributions

2.3.1 True voicing and primary nasals

Nasukawa (2005) and Backley & Nasukawa (2009) propose a typological argument linking true voicing (i.e. phonologically active phonetic voicing of obstruents in a language) to the presence of primary nasals (i.e. phonologically active nasal segments in contrast with obstruents). The core of the argument is an implicational asymmetry, where the presence of true voicing in a language implies the presence of primary nasals, but the mere presence of primary nasals does not imply the presence of true voicing. As shown in Table 1, they claim that all the possible combinations are attested except languages with true voicing and no primary nasals, which in turn implies that underlyingly, languages with true voicing are a subgroup of languages with nasals. This can be explained through the UVNH if the unified phonological feature primarily encodes nasality, so that true voicing can only be encoded in systems where that feature is active already for the encoding of nasality.

Table 1: Typology of true voicing and primary nasals according to Backley & Nasukawa (2009:68) and Breit (2017:16).

Backley & Nasukawa (2009)	**Breit (2017)**	**Primary Nasals**	**True Voicing**
Quileute	Quileute	✗	✗
Finish, English	Finish, English	✓	✗
(none)	Ditidaht, Rotokas Proper	✗	✓
Dutch, French, Thai	Dutch, French, Thai	✓	✓

However, Breit (2017) argues that the typological data relied on by Nasukawa (2005) and Backley & Nasukawa (2009) are incomplete, and that systems exist which fit into the apparent typological gap. Namely, Ditidaht (Wakashan) has no primary nasals but does have a series of true voiced plosives (Sylak-Glassman 2013:17). Incidentally, the current system in Ditidaht historically derives from the loss of primary nasals, which gave rise to the true voiced plosives, thus again showing a link between voicing and nasality but breaking the implicational asymmetry synchronically. As two further examples, Breit (2017) cites Rotokas Proper, which has true voiced plosives but no primary nasals (Firchow & Firchow 1969), and based on data from WALS (Maddieson 2013) and discussion in Sylak-Glassman, also Lushootseed (Salish), giving in sum at least three candidate systems to fill the purported gap. As discussed further for the case of Rotokas in Section 2.3.3, however, there is again evidence for a link between voicing and nasality in spite of the absence of primary nasals.

2.3.2 Ferguson's Universals

Ferguson (1966) proposes a number of universals concerning nasals and nasality. Of particular relevance to the voicing—nasality connection here are his universals 6-9, on what he calls 'secondary nasal consonants'. The term 'secondary nasal consonants' is used by Ferguson (1966) to refer to nasal consonantal phonemes which are not 'simple voiced nasals', and thus includes breathy, voiceless and/or aspirated nasals (see also Section 2.1.1), and glottalised/laryngealised nasals, palatalized nasals, emphatic nasals (e.g. in Arabic), prenasalised stops (see also Section 2.2.2), and nasalized clicks. Of these, the laryngeally modified nasals (i.e. breathy, voiceless, aspirated, glottalised/laryngealised/creaky nasals) are of particular relevance to the UVNH. Ferguson's Universals 6-8 imply that such laryngeally modified nasals are only found in languages which also have primary, modal voiced nasals (Universal 6); that they never exceed the number of primary, modal voiced nasals in the language (Universal 7); and that their frequency of occurrence is always lesser than that of the primary, modal voiced nasals in the language (Universal 8).

This asymmetry is phonologically unexpected if nasality and voicing are independent features which we expect to be equally distributable in principle (so that the combination [nasal, spread glottis] should be no more implausible than the combination [nasal, voice]). As discussed already in Section 2.1, there is no articulatory reason which would militate against either of these; the only link favouring a [nasal, voice] connection is the acoustic perceptibility of place cues in the transition from the nasal to adjacent segments. The acoustic perceptibility argument does not sufficiently explain the phonological asymmetry however, since a hypothetical language with a series of voiceless aspirated nasals but no modal voiced nasals could still satisfy the perceptibility simply by phonetically implementing all of these nasals with partial voicing, the same way this happens in all the attested languages in which we find both fortis and lenis nasal consonants. Breit (2017:18) argues, however, that the UVNH taken together with Laryngeal Realism (Harris 1994, Honeybone 2002) goes some way toward a phonological explanation for these universals, since the unified feature implies that the only possible representations for laryngeally modified nasals are those of a regular modal voiced nasal with another feature added to specify them as breathy, aspirated, creaky, etc. In other words, under these assumptions the modal voiced nasals are a strict featural subset of laryngeally modified nasals, so that it stands to reason that if the larger group of laryngeally modified nasals is present, the modal voiced variants must be allowable, too.

Ferguson's Universal 9 proffers that laryngeally modified nasals always arise diachronically from a sequence of a regular, modal voiced nasal and some other segment, the latter of which contributes the 'modified' quality. As an illustrative example for the (diachronically) derived nature of such nasals, consider the 'voiceless' (fortis) series of nasals in Icelandic, as shown in (14).

(14) Icelandic (Pétursson 1998; Bombien 2006)

[lampa]	'lamb'	[lam̥pɪ]	'lamp'
[hɛnta]	'to throw'	[hɛn̥ta]	'to be appropriate'
[hɛntɪ]	'hand'	[hɛn̥tɪ]	'to dispose of'

The fortis nasals in the right-hand column of (14) are typically found in positions where they precede what would otherwise be strongly preaspirated plosives so that it has been commonly argued that rather than being underlyingly (and at least diachronically) truly voiceless nasals, they result from coalescence of a modal nasal with the preaspiration of a subsequent segment (e.g. Haugen 1958, Árnason 1986), which also fits with analyses which propose that preaspirated stops in this position, where there is no nasal present, are underlyingly best captured as geminates (e.g. [saʰka] /sakka/ 'sinkstone'), as proposed e.g. by Thráinsson (1978), suggesting that the nasal here occupies the left-hand segmental slot of the geminate, still realising the fortisity/voicelessness that would be encoded in that position for a geminate voiceless plosive otherwise. This appears to be further corroborated by the fact that in northern dialects of Icelandic, which have postaspiration in this position, the nasals remain modal, as for instance with [hɛn̥tɪ] 'to dispose of', which is realised as [hɛntʰɪ] in such dialects (Bombien 2006:65).

While we are not aware of particular exceptions to the central proposition of Ferguson's Universal 9, there are cases where the link between laryngeally modified nasals and their diachronic precursors remains somewhat more tenuous. For example, there are also instances of initial fortis nasals in Icelandic, such as in [n̥iːta] 'to knot' (cf. lenis [niːta] 'to use'). The analogous argument to what has been proposed for the other fortis nasals in Icelandic suggests that these derive from (diachronic) underlying clusters, viz. /hniːta/, though as far as we know no strong comparative or historical arguments which would justify such a representation have been put forward yet, apart from simple analogy with the segments' distribution elsewhere in the language.

2.3.3 Rotokas

Ferguson's (1966) Universal 1 asserts that every language has at least one primary nasal consonant. Although Ferguson was already aware of Hockett's (1955:119) description of a few Salishan languages which appear to synchronically have no primary nasal consonants, there is a presumption that such systems have at least a diachronic precursor nasal which is proposed to have latterly given rise to voiced stops. Such diachronic developments are of course what would be expected under the UVNH, in that the shift from nasality to voicing (or vice-versa) would not represent a change in feature content, but merely in their configuration and/or phonetic interpretation. In addition to Salish, Section 2.3.1 briefly discussed a few more languages that synchronically appear to lack primary nasal consonants. Of these, as noted as early as Herbert (1986), Rotokas (Firchow & Firchow 1969) is a particularly illustrative case with respect to the UVNH. This is because there are two dialects, namely Rotokas Proper and Aita Rotokas, of which the former has no primary nasal consonants in its inventory, while the latter has no voiced obstruents in its inventory, as illustrated in (15).

(15) Consonant Inventories of two Rotokas dialects (Firchow & Firchow 1969)

a. Rotokas Proper

p	t	k
b	ɾ	g

b. Aita Rotokas

p	t	k
m	n	ŋ

This seems to suggest that voicing and nasality across the two dialects merely exhibit interpretational variants of the same underlying phonological entity, viz. a unified prime that is interpreted as voicing in Rotokas Proper and as nasality in Aita Rotokas. This is further underlined by the fact that the interpretation of the voiced/nasal series in Rotokas has been reported to be phonetically highly variable, i.e. allowing a large degree of ‘phonetic slack’, such that for instance /b/ might be realised variously as a voiced stop [b], a voiced fricative [β] or a voiced nasal [m]. This is illustrated further in example (16).

(16) Variable realization of voiced obstruents in Rotokas (Firchow & Firchow 1969: 274)

/βaβae/	[βaβa^{e} ~ baba^{e} ~ mama^{e}]	‘five’, ‘hand’
/ɾagai/	[ɾaga^{i} ~ daga^{i} ~ laga^{i} ~ naga^{i}]	‘I’, ‘me’
/βegei/	[βege^{i} ~ βeɣe^{i} ~ βeŋe^{i}]	‘we two’

This means that Rotokas Proper in effect is a system without primary nasals that shows optional nasalisation of voiced obstruents (Pirahã is another system of this type, see Everett 1986), which leads Firchow & Firchow (1969) to suggest what essentially has been described as the unified prime analysis above, namely that the two dialects simply reflect a differential preference for expressing the voicing/nasality category: voicing in Rotokas Proper, nasality in Aita Rotokas.

While this does not affect the demonstrative relevance of Rotokas for the UVNH, it is interesting to note that Robinson (2006), based on more recent fieldwork, proposes that Aita Rotokas (at least in his more recent sample) actually shows evidence for a voicing–nasality contrast. In particular, Robinson gives two minimal pairs, /buta/ ‘time’ v. /muta/ ‘taste, feel’, and /dito/ ‘hole’ v. /nito/ ‘remove embers’ to show that /b, m/ and /d, n/ are contrastive in Aita Rotokas. While Robinson reports that he was unable to identify a minimal pair for /g, ŋ/, it is suggested that this gap is due to his limited data rather than being representative of the system. Robinson suggests that a three-way contrast in Aita Rotokas compared to a two-way contrast in Rotokas Proper (which he refers to as Central Rotokas), along with ample evidence for a systematic correspondence between voicing and nasality across the two dialects, suggests a diachronic precursor with a three-way contrast, with nasality being diachronically lost in Rotokas Proper. Although not discussed further in Robinson (2006), this seems to suggest that voiced oral stops and voiced nasal stops merged diachronically in Rotokas Proper, which one might expect to go along with a potential loss of contrast. Unfortunately, Robinson did not further investigate whether such cases of presumed prior minimal pairs have led to any loss of contrast in Rotokas Proper, however he notably does report that one of the minimal pairs for /b, m/ in Aita Rotokas is a minimal pair for vowel length in Rotokas Proper, as shown in (17).

(17) Minimal pairs across Rotokas dialects (Robinson 2006)

a. Aita Rotokas		b. Rotokas Proper	
/buta/	‘time’	/buuta/	‘time’
/muta/	‘taste, feel’	/buta/	‘taste, feel’

The situation in (17) seems to suggest the possible involvement of displaced contrast, which leaves the question of diachronic directionality of developments more open. For instance, if Aita Rotokas was at one point indeed a two-way system as suggested by Firchow & Firchow (1969) and illustrated in (15a), but then developed a phonological contrast between nasality and voicing, this contrast might in cases such as (17) have obviated the significance of the length contrast (Tok Pisin would also present itself as a possible source domain for the development of such a contrast, given widespread bilingualism). This certainly is a case where more extensive, detailed data could lead to interesting insight into the synchronic and diachronic development and interaction of voicing and nasality, particularly in the formation of these types of rare systems that either lack primary nasal consonants or voiced oral obstruents.

3 Analyses and theoretical proposals

3.1 Head/dependency in Element Theory

To encapsulate the robust connection between nasality and true voicing within the context of Element Theory, Nasukawa (1998), Ploch (1999) and others suggest consolidating consonantal nasality (represented by |N|) and true obstruent voicing (represented by |L|) into a unified nasal-voice category. While Nasukawa originally labelled the unified element |N|, others (e.g. Kula & Marten 1998; Ploch 1999; Breit 2013, 2017) have adopted the now-common label |L| for the unified element (as we also do here), though it ought to be emphasised that this is merely a notational matter with no theoretical weight behind it. This unification approach presents distinct benefits, highlighting the advantages of considering these two properties as distinct phonetic expressions of a single representational category.

While researchers working in Element Theory commonly affirm the UVNH and have adopted the unified element approach to voicing and nasality, there remains a disputed point on how the two phonetic properties are encoded by |L|. On the one hand, Nasukawa (1998, 2005) and Ploch (1999), propose that nasality is the phonetically realized form of non-headed |L| in a given phonological expression while headed $|\underline{L}|$ (the underline denotes headedness) manifests itself as true voicing, as illustrated in (15a).[6]

(15) Non-headed |L| and headed |L| and their phonetic manifestations

		non-headed |L| (*fundamental*)	headed $\|\underline{L}\|$
a.	Nasukawa (1998, 2005b) Ploch (1999)	nasality	true voicing
b.	Kula & Marten (1998) Breit (2013, 2017)	voicing	nasality

Conversely, Kula & Marten and Breit (2013, 2017) have proposed that that non-headed |L| encodes voicing and headed $|\underline{L}|$ encodes nasality, as shown in (15b).

[6] In light of the current discussion on the correlation between nasality and obstruent (true) voicing, the phonetic realizations of |L| in vowels/nuclei lie beyond the scope of our current discussion. Nevertheless, it is pertinent to briefly outline two perspectives under the claim that non-headed |L| and headed |L| are phonetically realized as nasality and true voicing, respectively.
Ploch (1999) posits that, contrary to non-nuclear positions, the non-headed |L| and headed |L| within nuclei should be phonetically realized as low tone and nasality, respectively. Conversely, Nasukawa & Backley (2018) assert, akin to non-nuclear positions, that the non-headed |L| and headed |L| within nuclei should be phonetically realized as nasality and low tone, respectively.
There are two principal arguments against Ploch's view and in support of Nasukawa & Backley's view. First, in languages exhibiting nasal harmony, nasality in Cs typically spreads to adjacent Vs. If, as proposed by Ploch (1999), nasality in Cs and Vs represents the realized forms of non-headed |L| and headed |L| respectively, a mechanism is required to invert the headedness of |L| when it spreads from C to V. On the other hand, according to Nasukawa & Backley (2018), such an operation is unnecessary, as nasality serves as the phonetic manifestation of |L| in both Cs and Vs.
Second, the perspective suggesting that the headed |L| is realized as a low tone in nuclei finds support in tonogenesis (Hyman 1979, among others), which explains the strong correlation between obstruent voicing and vocalic low tone: the latter evolves from the former in diachronic terms.

A pivotal distinction between (15a) and (15b) lies in determining the foundational structure of |L|: proponents of (15a) consider nasality to be the basic exponent of the fundamental (non-headed) representation with voicing being the exponent of a more complex (headed) representation. Proponents of (15b) argue for the opposite, i.e. that voicing expones the fundamental, non-headed representation while nasality expones a more complex, headed representation. Each standpoint finds justification in distinct rationales. The supporting evidence is categorized into two types: phonological and phonetic.

In the view presented in (15a), as per Nasukawa (1998, 2005), the assignment of head status to true voicing rather than to nasality can be justified in at least three phonological respects. First, the representations in (15a) can encode an implicational universal between voicing and nasality. As discussed in Section 2.3.1, Nasukawa (2005) and Backley & Nasukawa (2009) argue that there are languages that feature neutral/spontaneous voicing[7] without nasals, while there are no systems displaying truly-voiced plosives without also having nasals. If this observation is correct, it leads to the inference that the existence of true voicing (which is not basic) implies the existence of nasality (which is basic). Such an implicational universal is straightforwardly captured by the representations in (15a).

Second, the representations in (15a) are argued to capture the 'discretionary character' of true voicing. Contrastive nasality is employed by nearly all languages, while the use of true voicing (in contrast to neutral or spontaneous voicing) appears to be subject to parametric variation. This optional status of true voicing reflects the non-essential aspect of headedness: certain systems permit the heading of |L|, while others exclude that structural possibility.

Third, the representations in (15a) delineate distinctions in complexity between true voicing and nasality. In Nasukawa's (1999) analysis of prenasalization and velar nasalization, nasality is deemed to be structurally more fundamental than true voicing. This analysis stems from instances where true voicing is suppressed in intervocalic contexts, giving rise to nasality in its place, as may be observed in certain dialects of Japanese, various Western Indonesian languages, and several Bantu languages. If nasality is more fundamental, this can be seen as a reduction in melodic complexity in line with the perspective advanced by Harris (1994, 1997) who asserts that segmental structure exhibits less complexity in weak positions compared to strong positions.

Kula & Marten (1998) and Breit (2013, 2017), on the other hand, propose encoding the nasality–voicing contrast the other way around. They advance two principal arguments against the view in (15a).

Breit (2017) first points out that the typological facts discussed by Nasukawa (2005) and Backley & Nasukawa (2009) are a tendency rather than absolute. He claims that there are not many but a few languages such as Ditidath (Wakashan: Sylak-Glassman 2013ab), Lushootseed (Salish: Maddieson 2013) and one dialectal variant of Rotokas (Firchow & Firchow 1969), which have voicing but no nasality. The existence of such languages would undermine the claim that the presence of nasality is a prerequisite for true voicing. Given this, he argues that there is no implicational universal that could serve as supporting evidence for determining the headed/non-headed status of |L| for nasality and voicing, instead proposing that the typological distribution is compatible with either structural configuration.

[7] In Nasukawa & Backley (2018), a nucleus itself is phonetically realized as the carrier signal of the speech sound, which is schwa-like and phonetically voiced. It also affects its prosodically-dependent onsets: an onset with no |L| is phonetically realized as spontaneously voiced, where the segment is interpreted as 'voiced' in aspiration languages, while it is interpreted as 'voiceless' in voicing languages.

Second, Breit (2017) claims that Backley & Nasukawa's assumption that the existence of a non-headed element is a prerequisite for its headed counterpart, is not generally a well-established assumption in standard ET (and Government Phonology, more widely). Generally, the theory does not have a prerequisite requirement on headedness assignment and a given phonological system is free to employ a headed element without making use of its non-headed counterpart (in fact, a language may have licensing constraints forcing an element to only occur as head, see e.g. Breit 2023). This argument again aims more at weakening support for the view that voicing and nasality are universally encoded as in (15a), rather than being construed as an argument for (15b).

Given these points, Breit (2013, 2017) questions the assumption (15a) made by Nasukawa (1997, 1998, 2000), Ploch (1999), and Backley & Nasukawa (2009). He claims that the nasality–voicing contrast could plausibly be encoded the other way around, i.e, headed |L| is realized as nasality and non-headed |L| as voicing. He argues that this 'alternative implementation' is supported by the idea that headedness may encode additional properties, for instance to increase salience in terms of sonority, at phonetic interpretation while at the same time he views it as unlikely that a headed element could 'lose' a phonetic/acoustic property that it expones in a non-headed configuration. Since nasals are generally more sonorous than voiced non-nasals, and nasals appear to be phonetically inherently voiced by default, he argues that the assumption that headed $|\underline{L}|$ is interpreted as nasality and non-headed |L| as voicing is most consistent with the general interpretive behaviour of headed vs non-headed configurations at the phonetic interface.

In addition to discussing the phonetic realization of headedness, Breit (2017) further discusses the phonological strength of nasality. According to him, nasality is phonologically stronger than voicing since the former is rarely subject to lenition and deletion cross-linguistically and there are instances where nasality might plausibly be analysed as fortition (e.g. Iverson & Salmons 1996). Given that headedness is deemed to be phonologically strong, it is argued that nasality should be the phonetically interpreted form of headed $|\underline{L}|$, while voicing, being phonologically weaker than nasality, should be the realization of non-headed |L|.

Both approaches (15a) and (15b) offer intriguing perspectives to the ongoing debate, prompting an exploration into which offers a more robust analysis of recurring phonological phenomena. Alternatively, we may consider accepting the structures for nasality and obstruent voicing as being parametric variants, where the choice between them is made on a language-by-language basis. This could hinge upon the working hypothesis chosen to guide our investigation into the direction of this research inquiry.

3.2 Dependency Phonology

The UVNH has also been developed in Dependency Phonology (see the chapter *Dependency Relations between Phonological Units*; Anderson & Ewen 1980, 1987; van der Hulst & van de Weijer 2018), an approach closely related to Government Phonology and Element Theory. Like the ET approach in Section 3.1, Botma (2004, 2009) and Botma & Smith (2007) propose a single element |L| responsible for both voicing and nasality. In contrast to the ET proposals' reliance on headedness, they propose that dependent |L| is interpreted as voicing when it is linked to an obstruent manner component, and as nasality when it is linked to a sonorant manner component.

Following Humbert (1995) and Kehrein (2002), they assume that segments are organised as shown in (16):[8]

(16)

```
O/N/C
 |‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾\
manner             phonation
 |
place
```

In (16), O/N/C are categorizers for Onsets, Nuclei and Codas, respectively, which dominate a manner component and a phonation component, both specified by the elements {ʔ, H, L}, and manner in turn dominates a place component specified by the elements {A, I, U}. In this system, stops will have |ʔ| in the manner component, fricatives |H|, and sonorants |L|, the latter two consistent with the fact that sonorants are by default voiced/lenis while fricatives are by default voiceless/fortis. Phonation is a marked component which can carry material further specifying the realisation of a segment (or in fact a larger domain it is attached to).

Let us first consider the role of |L| in the manner component, where it is responsible for the encoding of segments' sonorancy. As shown in (17a,b,d) this essentially encodes a sonorant/obstruent contrast. But |L| in the manner component can also co-occur with other manner specifications. In co-occurrence with |ʔ|, it specifies a sonorant obstruent, the category under which this approach classifies primary nasal consonants such as the /m/ in (17c). In co-occurrence with |H|, sonorants with obstruent properties can be encoded, as shown for /r/ in (17e). The representation of nasals in (17c) is justified in part by evidence that nasals frequently pattern with other sonorants and thus should be encoded as such, and in part by the analysis of nasal-glide alternations as linking/delinking of |ʔ| from segments with a structure such as (17c) (see Botma 2004:76—83).

(17)

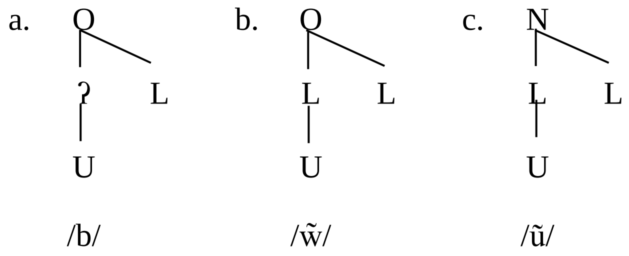
a.	b.	c.	d.	e.
O	O	O	O	O
|	|	|	|	|
L	ʔ	L	H	H
|	|	|		|
I	U	ʔ		L
		|		|
		U		I
/j/	/p/	/m/	/h/	/r/

As with the Element Theory approach, a representation such as (17c) incorporates some phonetic slack in terms of its interpretation. On the one hand, this captures the expectation that 'sonorant stops' will be voiced by default. On the other hand, it manifests directly in languages where voicing and nasality share a phonological category, such as Rotokas (Section 2.3.3). In Aita Rotokas, (17c) is phonetically interpreted as /m/ by default, while Rotokas Proper interprets the same structure as /b/ by default — but both incorporate the necessary phonetic slack to allow the phonetic voicing and nasality categories to be employed for paralinguistic and emphatic functions, and to maintain mutually compatible phonological representations (Botma 2004:64). This approach to languages where a phonetic voicing—nasality distinction

[8] Botma (2004) and Botma & Smith (2007) differ in whether they assume an x-slot intervening between O/N/C and manner. For simplicity we omit the x-slots here.

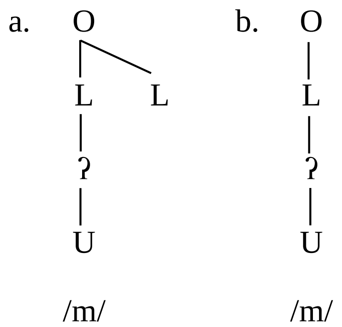
is not phonologically contrastive is commensurate with analyses in classical Element Theory (see e.g. Breit 2017).

The contextual aspect of the interpretation of |L| in Dependency Phonology enters the picture when |L| is present in the phonation component. Here |L| is interpreted as voicing on segments with an obstruent manner component, as in (18a), but as nasality on segments with a sonorant manner component, as in (18b-c).

(18)


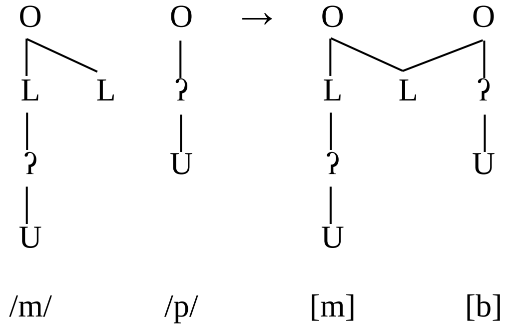
Since the phonation component is representationally more marked than the manner component, representations such as (18a) also lend themselves to explanations of typological generalisations regarding the default phonation of different types of segments: for obstruents, the plain voiceless version in (17b) is unmarked while the voiced one in (18a) is marked. Conversely, for sonorants the voiced version is unmarked by virtue of containing |L| in the manner component (a voiceless version will specify |H| in the phonation component).

Because |L| in the phonation component of obstruent-manner segments is interpreted as voicing, primary nasal consonants must be encoded with |L| in the manner component (as in 17c), so that they fit into the category of ‘sonorant stops’. An |L| in the phonation component of such a sonorant stop will then be interpreted as nasality, as shown in (19a). This leads to two possible representations for primary nasal consonants which may co-exist in a language, i.e. (19a) and (19b):

(19)


While the representations in (19a) and (19b) may receive the same phonetic interpretation, they are phonologically distinct, and Botma (2004) argues that this distinction is borne out in their phonological activity. The |L| of the nasal in (19b) is phonologically inert, but because of its presence in the phonation component, the nasalising |L| in (19a) is phonologically active. Consider for instance the postnasal voicing processes in Puyo-Pongo Kichwa and Yamato Japanese illustrated in (3) and (4) in Section 2.2.1. These can be captured as cross-segmental linking of the phonation component, as illustrated in (20).

(20)



In the resultant representation of (20), the shared |L| is interpreted as nasality on the sonorant stop and as voicing on the following obstruent. One strength of this analysis is that phenomena such as progressive voicing and regressive nasalisation can be captured in the same way, since the |L| in the phonation component is available for cross-segmental linking generally, with the language’s phonology having to simply determine the environments under which linking does or does not take place.

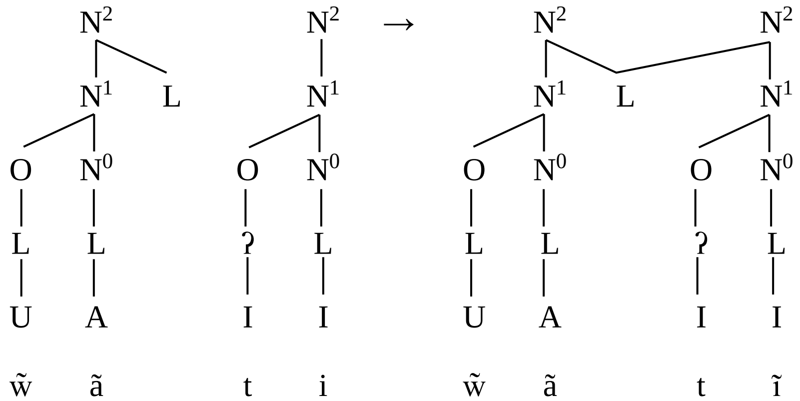
Nasal harmony is well known to target different domains, e.g. syllables, morphemes, phonological words (see the chapter *Nasal Harmony*). The dependency approach of Botma (2004, 2009) and Botma & Smith (2007) captures this in an analogous way to the segment-adjacent sharing in (20), with the site of attachment and linking in those cases involving a higher, suprasegmental projection (see also Kehrein 2002). Consider for instance the data in (21), showing some aspects of nasal harmony in Southern Barasano.

(21) Southern Barasano harmony (Smith & Smith 1971, cited in Botma & Smith 2007:43).

kãmõka	‘rattle’	w̃ãtĩ	‘demon’	hati-ãmĩ	‘he sneezes’
mãsã	‘people’	rimã	‘poison’	hiamãkõnõ	‘ten’

The data illustrate that nasality in Southern Barasano spreads rightward across the word domain, and that fortis obstruents remain unaffected while sonorants and lenis obstruents (viz. [m, n] which alternate with [b, d]) are subject to nasalisation. Botma (2004:140) and Botma & Smith (2007:43) analyse this as rightward spreading of an |L| phonation component attached at the syllabic N^2 projection (N'' in Botma 2004, with additional |L| at the O^0/N^0 tier not shown here). This is illustrated in (22) for [w̃ãtĩ] ‘demon’:

(22)



To account for the fact that fortis obstruents do not undergo voicing, Botma (2004:140) proposes the *Consistency of Dependency Relations* principle, which essentially mandates that |L| shared at a higher projection is only interpreted on segments of the same category. This means that the /t/ in (22) does not interpret the |L| attached at N^2, because unlike the originator it is not specified as a sonorant (which would require |L| in the manner component). This principle also links back directly to Botma's proposed category of 'sonorant stops', in that alternation between [b, d, g] and [m, n, ŋ] relies on assuming that [b, d, g] are underlyingly specified as sonorant stops (although being interpreted as oral in the absence of a phonation component specified |L|), so that they are subject to the interpretation of a high-attached |L|. Were they instead underlyingly specified as in (18a), we would expect them not to participate in the nasal harmony process.

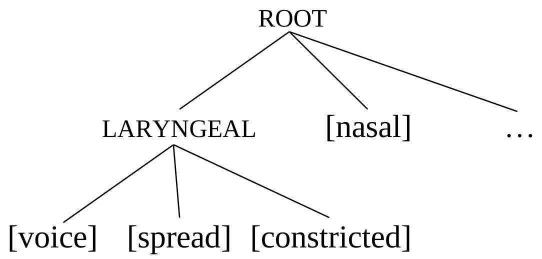
The two principal factors which set the Dependency Phonology analysis apart from the Element Theory analyses in Section 3.1 are (i) the rich differentiation that can be given to base segments (e.g. contrasting sonorant stops and obstruent stops, active and inert phonation components) and (ii) the unified process by which voicing and nasality are distributed both among adjacent segments and throughout larger domains, with the projection of linking being a principal factor distinguishing assimilatory from harmonic processes. In contrast, Element Theory with a single head/dependent distinction cannot utilise |L| for more than three distinctions of basic categories, which are always already directly tied to the interpretation of |L|, and spreading processes are not as uniform as they have to directly take into account whether |L| is headed or unheaded.

3.3 Feature Theory

To the best of our knowledge, the UVNH has not been explored to any significant extent in Feature Theory, which we understand to loosely encompass theories of phonological features/primitives recognisably descendant from the approaches in Jakobson, Fant & Halle (1952), Jakobson & Halle (1956) and Chomsky & Halle's (1968) *Sound Pattern of English*. In part, this may be due to the common assumption of features that are more closely tied to articulatory substance than is the case with Element Theory and related approaches. However, many conceptions of features in the *Sound Pattern of English* tradition also incorporate acoustic and aerodynamic aspects or some degree of phonetic slack, so that it is difficult to say per-se that the difference in articulators involved (the velum vs. the vocal folds) rules out a unified feature in these theories. This is of course particularly true for emergent substance free approaches to phonological features (see the chapter *Substance-free Approaches to Phonology*; Scheer 2019; Boersma, Chládková & Benders 2022), which would appear to us to be a prime candidate for exploring the UVNH in a non-element-theoretic framework (for further discussion of the nature of features and the commensurability of different approaches see e.g. van 't Veer et al. 2023). Aside from the possibility of directly encoding a unified [nasal-voice] prime in substance-free approaches, the advent of Autosegmental Phonology (Goldsmith 1976) and Feature Geometry (Clements 1985) have also brought with them the potential to encode aspects of the UVNH in terms of differential attachment sites and hierarchical grouping.

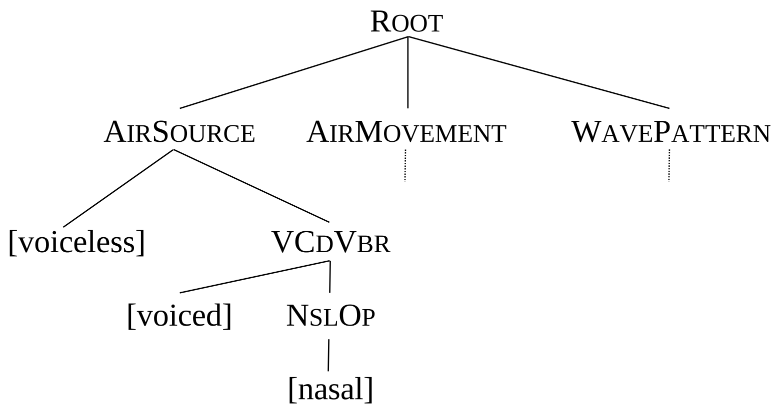
Most accounts of Feature Geometry place [nasal] either under the MANNER node (possibly itself contained in a SUPRALARYNGEAL branch or similar) while placing [voice] and related features under a LARYNGEAL node (e.g. Clements 1985; Yip 1989; Keyser & Stevens 1994), or they place [nasal] directly under the ROOT node, typically as a sister to a LARYNGEAL branch which in turn dominates features such as [voice], [spread], and [constricted] (e.g. McCarthy 1988; Halle 1995; Halle, Vaux & Wolfe 2000; Lahiri & Kotzor 2003), as shown in (23).

(23) A common feature-geometric arrangement



While the root attachment of [nasal] may lend itself to capturing some of the distributional and procedural aspects of nasality, and the laryngeal grouping is articulatorily well-grounded, neither of these configurations encapsulate in any way the generalisations across voicing and nasality that the UVNH seeks to capture. An exception to this is Kuroda's (2008) Aerodynamic Feature Geometry (ADFG), a version of feature geometry based on the aerodynamic design of the articulatory organ.

(24) Aerodynamic Feature Geometry (Kuroda 2008)



In this model, obstruent voicing and sonorant voicing correspond to the feature [voiced] and the node VCDVB respectively. The strong correlation between voicing and nasality is captured by the node VCDVBR, which dominates both: [voiced] is directly dominated by VCDVBR, while [nasal] indirectly dominated by it via NSLOP, as illustrated in (24).

3.4 Discussion

The discussion above highlights both divergence and potential convergence in how phonological theories treat nasality and voicing, particularly in light of UVNH. While mainstream Feature Geometry (FG) and its descendants (e.g. Clements 1985; McCarthy 1988; Halle 1995) typically represent [voice] and [nasal] in distinct subtrees—articulatorily grounded in the laryngeal and supralaryngeal systems, respectively—Kuroda's (2008) ADFG departs from this tradition by offering a model in which nasality and voicing are grouped under a shared aerodynamic mechanism. In ADFG, both features are subordinated under the node VCDVBR, which captures their tendency to co-occur and interact in natural language.

However, compared with Element Theory, which directly encodes the correlation between nasality and voicing by treating them as two phonetic manifestations of a single element, ADFG maintains a more complex structural distinction. In ADFG, nasality is represented either by [nasal] or by the node NSLOP, while obstruent voicing is assigned the feature [voice], and sonorant voicing is associated with VCDVBR itself. Thus, while ADFG offers a unified aerodynamic grounding, the mapping between feature and function remains somewhat intricate.

Despite these structural differences, many feature-based theories implicitly acknowledge a functional or acoustic affinity between voicing and nasality. For example, even in models where [nasal] and [voice] are structurally distant—such as when [nasal] is attached directly to the ROOT node and [voice] to a LARYNGEAL node—they may still interact within phonological rules or constraints. Furthermore, although many classical models are grounded in articulatory anatomy, they often incorporate acoustic or aerodynamic factors to some degree, as noted in foundational works and more recent elaborations.

What all theories engaging with the UVNH have in common is a recognition—whether explicit or implicit—of a natural affinity between nasality and voicing as phonological properties. Where they differ is in how they conceptualize and formally represent this relationship:

- FG typically separates the two based on articulatory loci.
- ADFG unifies them under shared aerodynamic processes, with both features dominated by the node VCDVBR.
- ET encodes nasality and voicing using a single element, thereby capturing their unity in a maximally economical fashion.

Thus, while the surface architectures vary—whether in terms of nodes, tiers, or elemental representations—many frameworks converge on similar functional generalizations through different theoretical pathways. This convergence suggests that the connection between nasality and voicing is robust across phonological models.

Such common ground opens the door to broader cross-theoretical exploration. Rather than viewing these models as mutually exclusive, it may be fruitful to investigate how insights from one framework—such as the aerodynamic grounding of ADFG or the representational minimalism of ET—can inform or refine others. Bringing the UVNH into the conceptual space of models that have traditionally avoided such unification may yield novel insights, not only for theoretical cohesion but also for our understanding of the phonetics-phonology interface. In this way, exploring the shared implications of voicing and nasality across models could contribute meaningfully to the development of phonological theory as a whole.

4 Perspectives

The Unified Voicing and Nasality Hypothesis has been extensively investigated and evaluated within Element Theory/Government Phonology and some adjacent frameworks over the last few decades, where a unified voicing-nasality feature is now an established part of the canon. Even so, particularly outside of Element Theory, the UVNH continues to present a wider case for rethinking the traditional approach of representing voicing and nasality as distinct and independent phonological features and it is likely that exploration of similar approaches in different frameworks could lead to further insights on the phonological representation of voicing and nasality. Particularly proposals employing a singular unified prime have interesting implications for the mapping between phonetic categories and the cognitive reality of their phonological counterparts, suggesting that though phonetically distinct, voicing and nasality may reflect two sides of a single cognitive category. While the data discussed has much to offer with respect to the categorisation, interrelation and contrastiveness of the voicing and nasality, the question of category formation and more generally data from acquisition and learnability has not yet been brought to bear on the issue and could provide further insights both on the cognitive reality of a unified feature and the implementational possibilities to encode the

UVNH more broadly. Finally, there also remain fruitful empirical pastures for further exploration of the UVNH, for instance phenomena such as nasal shielding (Dobui et al. 2024), nasal fortition vis-à-vis nasality as 'hypervoicing' (Iverson & Salmons 1996), the flip-flopping behaviour of nasality and voicing in some systems (Bedar and Chergui 2024).

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See Also

- *The Atoms of Phonological Representations*
- *Partially Nasal Segments*
- *Dependency Relations between Phonological Units*
- *Elements for Segmental Contrasts*
- *Laryngeal Contrasts*